

MA.912.AR.1.3

Benchmark

MA.912.AR.1.3 Add, subtract and multiply polynomial expressions with rational number coefficients.

Benchmark Clarifications:

Clarification 1: Instruction includes an understanding that when any of these operations are performed with polynomials the result is also a polynomial.

Clarification 2: Within the Algebra I course, polynomial expressions are limited to 3 or fewer terms.

Connecting Benchmarks/Horizontal Alignment

- MA.912.NSO.1.1

Terms from the K-12 Glossary

- Polynomial

Vertical Alignment

Previous Benchmarks

- MA.7.AR.1.1
- MA.8.AR.1.2
- MA.8.AR.1.3

Next Benchmarks

- MA.912.AR.1.5, MA.912.AR.1.6, MA.912.AR.1.7
- MA.912.AR.6.3

Purpose and Instructional Strategies

In middle grades, students added, subtracted and multiplied linear expressions. In Algebra I, students perform operations on polynomials limited to 3 or fewer terms. In later courses, students will perform operations on all polynomials.

- Instruction includes making the connection to dividing a polynomial by a monomial and the understanding that division does not have closure.
- Reinforce like terms during instruction (using different colors can be a strategy to help identify them as unique from one another).
- Instruction includes the use of manipulatives, like algebra tiles, and various strategies, like the area model, properties of exponents and the distributive property.

- Area model

The expression $(2x^2 + 1.5x + 6)(3x + 4.2)$ is equivalent to $6x^3 + 12.9x^2 + 24.3x + 25.2$ and can be modeled below.

	$2x^2$	$1.5x$	6
$3x$	$6x^3$	$4.5x^2$	$18x$
4.2	$8.4x^2$	$6.3x$	25.2

- Instruction should not rely upon the use of tricks or acronyms, like FOIL.
- Although within the Algebra I course, polynomial expressions are limited to 3 or fewer terms, this restriction only refers to the expressions given to the student, not the expression after the operation applied.

Common Misconceptions or Errors

- Students may not understand the meaning of closure or the operations it applies to with polynomials.
- Students may not understand like terms or the properties of exponents.

Strategies to Support Tiered Instruction

- Instruction includes the use of shapes or colors to demonstrate like terms. Teacher must ensure that students understand that the sign in front of the terms are key when combining like terms.

- For example, different colors can be used when adding the polynomials shown below.

$$\begin{aligned}(x^2 - 4x + 12) + (3x^2 + 7x - 4) \\ (x^2 - 4x + 12) + (3x^2 + 7x - 4) \\ x^2 + 3x^2 - 4x + 7x + 12 - 4 \\ 4x^2 + 3x + 8\end{aligned}$$

- Teacher provides examples showing that polynomials are closed under the operations of addition, subtraction and multiplication, but not under division.
- For example, when subtracting or multiplying polynomials (as shown below), the result is always a polynomial.

$$\begin{aligned}7x + 5 - (3x + 8) &= 4x - 3 \\ (7x + 5)(3x + 8) &= 21x^2 + 71x + 40\end{aligned}$$

- For example, when dividing polynomials (as shown below), the result may or may not be a polynomial.

$$\begin{aligned}\frac{2x^2+5x}{x} &= 2x + 5 \text{ (which is a polynomial)} \\ \frac{2x^2+5x+1}{x} &= 2x + 5 + \frac{1}{x} \text{ (which is not a polynomial)}\end{aligned}$$

Instructional Tasks

Instructional Task 1 (MTR.3.1, MTR.4.1)

- Part A. Determine the sum of $3x^2 - 2x + 5$ and $\frac{1}{6}x^2 + 7x + \frac{8}{7}$. Explain the method used in determining the sum.
- Part B. Discuss whether the addition of polynomials will always result in another polynomial. Why or why not?
- Part C. Determine the difference of $3x^3 - 2x^2 + 5$ and $x^2 - 0.25x + 1.24$. Explain the method used in determining the difference.
- Part D. Discuss whether the subtraction of polynomials will always result in another polynomial. Why or why not?
- Part E. Determine the product of $2x + 5$ and $\frac{2}{9}x^2 - \frac{11}{2}x + 1$. Explain the method used in determining the product.
- Part F. Discuss whether the multiplication of polynomials will always result in another polynomial. Why or why not?
- Part G. Determine the quotient of $9x^2 - 3x + 12$ and $3x$. Explain the method used in determining the quotient.
- Part H. Discuss whether the division of polynomials will always result in another polynomial. Why or why not?

Instructional Items

Instructional Item 1

Determine the sum of the expression $\left(\frac{3}{4}x^3 - \frac{2}{3}\right) + \left(-\frac{1}{2}x^2 + x + \frac{5}{6}\right)$.

Instructional Item 2

Determine the value of the x^2 term when the expression $\left(x^2 + \frac{3}{4}x - \frac{1}{2}\right)$ is multiplied by $\left(x - \frac{2}{3}\right)$.

Instructional Item 3

Determine the difference of the expression $(-0.4x + 0.5x^2 + 2) - (0.6 + x^2 + 0.5x)$.

**The strategies, tasks and items included in the B1G-M are examples and should not be considered comprehensive.*